AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 hours

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Instructions:

- Total Marks: **50.**
- This exam consists of 5 questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are .... allowed.

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You should answer all the questions, giving 50 possible marks. Show your work.

Q1. [1+3+3+ 3 Marks]

a) Given the function f(x) = $(x-1)^2$ -2

i) Find the domain of the function f.

ii) Write the vertex and axis of symmetry.

iii) Find the zeros for the quadratic function f(x).

b) Use synthetic division *to determie if* $x-5$ *is a factor of the polynomia* $f(x) = x^4 - 5x^3 + 2x - 5$.

Q2. [3+4+3 marks]

a) Find the slope and y-intercept of the line 8x -6y =5

b) Given the function f(x) =x^3 +1, find the inverse of f(x).

c) If w is inversely proportional to the square of v, and w=3 when v=6, find w when v=3.

Q3. [4+2+4marks]

a) Solve the inequality: $x^2 - 1 < x + 5$

b) Find the domain of the following and write the answer in interval notation

i) F(x)= 2 e^{x-1} +3

ii) G(x)= $\sqrt{4x+5}$

Q4: [3+3+4 marks]

Solve for x:

a) e^{-2x} =0.2

b) $\ln x + \ln(x+3) = \ln 10$

c) $\log_5(3x+1) = 2$

Q5. [2+3+3+2 marks]

a. The standard configuration for license plate is 3 digits followed by 3 letters. How many different license plates are possible if digits and letters are repeated.

b. Find the indicated probability. P(A)=$\frac{1}{3}$, p(B)= $\frac{1}{4}$ and p(A∪ B) = $\frac{5}{12}$

i) P(A ∩ B)

ii) P(B | A)

iii) Are the two events A and B independent. Justify your answer.

End of questions

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